试 题 答 案

1. Single choice（10%, 2% for each）

(1) C (2) A (3) B (4) C (5)A

2. True of False（10%, 2% for each）

(1) T (2) F (3) F (4) F (5)T

3. (10%)

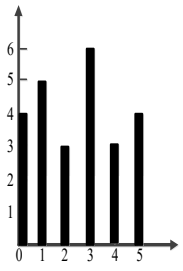
Nearest neighbor interpolation: P1=9 ,P2=2; (4%)

bilinear interpolation:

P1=7 (3%)

P2=4 (3%)

4. (10%)



(3%)

灰度值	0	1	2	3	4	5
像素个数	4	5	3	6	3	4
出现概率	0.16	0.2	0.12	0.24	0.12	0.16
累计概率	0.16	0.36	0.48	0.72	0.84	1
均衡化映射	1	2	2	4	4	5

(5%)

2	4	2	2	4
2	1	1	1	5
4	4	1	4	5

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4	2	2	5	4
2	4	2	5	4

(2%)

5.(8%)

Prove:

$$F(f(x,y)e^{j2\pi(u_0x/M+v_0y/N)})$$

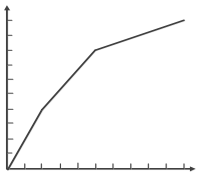
$$=\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{j2\pi(u_0x/M+v_0y/N)}e^{-j2\pi(ux/M+vy/N)} \quad (4\%)$$

$$=\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi((u-u_0)x/M+(v-v_0)y/N)} \quad (4\%)$$

So, $f(x,y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u-u_0,v-v_0)$

6. (8%)

$$f(x)=\begin{cases} 2x & 0\le x<20 \\ \frac{4}{3}x+\frac{40}{3} & 20\le x<50 \\ \frac{2}{5}x+60 & 50\le x<100 \end{cases} \quad (6\%)$$



(2%)

7. (10%)

4	12	2	6
10	7	7	8
6	7	8	6
6	12	4	10

The results of Average filter(2%)

4	12	2	6

10	8	7	8
6	8	8	6
6	12	4	10

The results of Midpoint filter (4%)

4	12	2	6
10	6	6	8
6	6	8	6
6	12	4	10

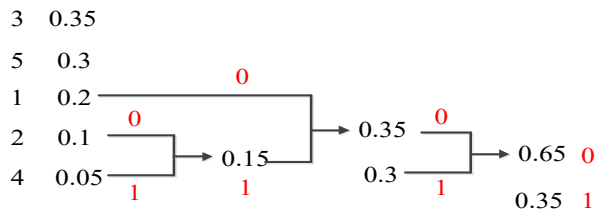
The results of Median filter (4%)

8.(12%)

(a)

灰度值	1	2	3	4	5
像素个数	4	2	7	1	6
出现概率	0.2	0.1	0.35	0.05	0.3

(2%)



1:000 2:0010 3:1 4:0111 5:01 (7%; 4% for coding steps, 3% for code)

$L_{avg} = 2.15$ C=8/2.15=3.72 R=1-1/C=0.73 (3%)

9.(10%)

(1) $A \circ B = (A \ominus B) \oplus B$ (2%)

(2)

0	0	0	0	0	0
0	0	1	1	0	0
0	0	0	1	0	0
0	0	1	1	0	0
0	0	0	1	0	0
0	0	0	0	0	0

(4%)

0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	1	0
0	1	1	1	1	0
0	0	1	1	1	0
0	0	0	1	0	0

(4%)

10.(6%)

(a) Get the size of image

(b) Initialize the variable segresults

(c) Set the value of regresults(row,col) to be 2. (2% each)

11.(6%)

Include capture image, image pre-process, feature extraction, similarity measure and recognition (3%)

Details (3%) 实现方式多样，酌情给分。